



Step-by-Step: BigQuery in action

This reading provides you with the steps the instructor performs in the following video, [BigQuery in action](#). The video focuses on how to create a query to view a small section of data from a large dataset.

Keep this guide open as you watch the video. It can serve as a helpful reference if you need additional context or clarification while following the video steps. This is not a graded activity, but you can complete these steps to practice the skills demonstrated in the video.

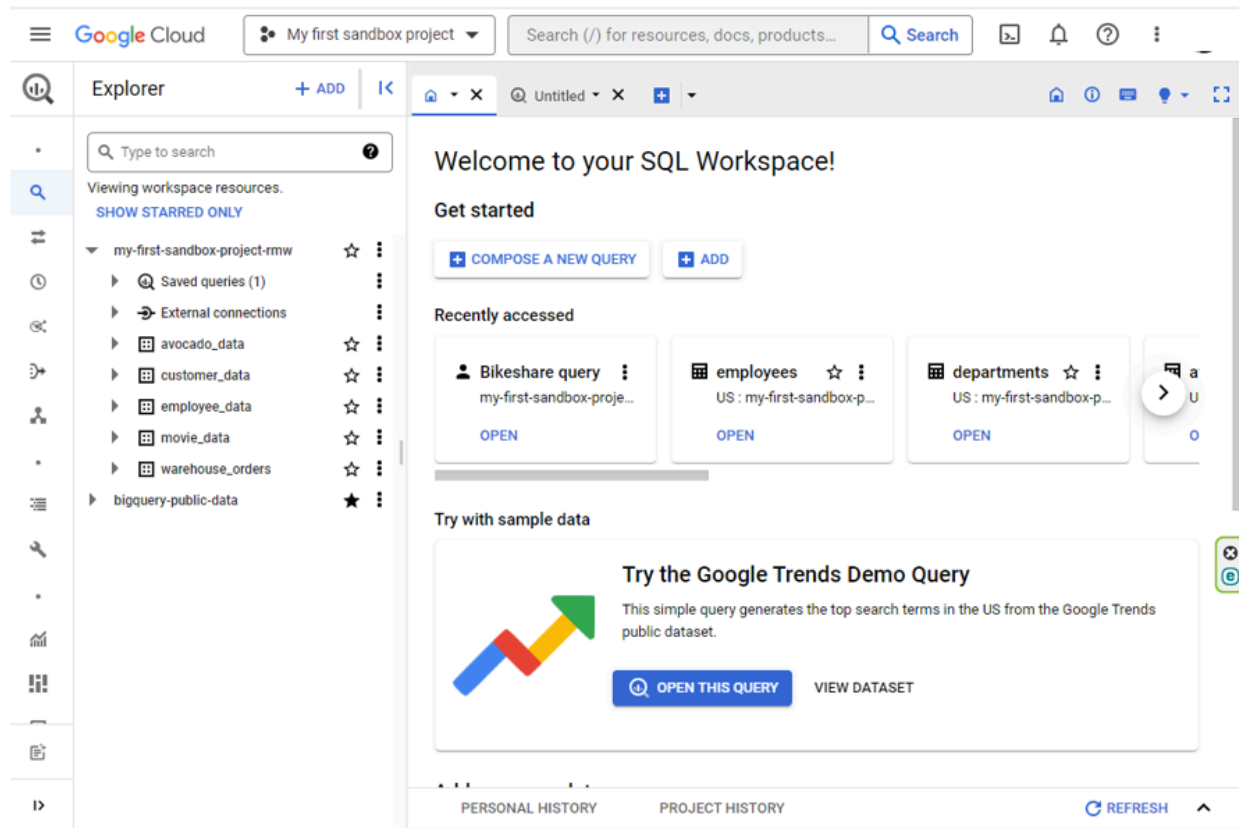
What you'll need

To follow along with the examples in this video, log in to your BigQuery account and follow the instructions to star bigquery-public-data in The Explorer pane section of the previous reading, [Get Started with BigQuery](#).

Example 1: Preview a section from a table viewer

A database is a collection of data stored in a computer system. Query languages such as SQL enable communication between databases and data analysts. You discovered earlier that a relational database is made up of several tables that may be joined together to create relationships. Primary and foreign keys serve as representations of these relationships. To extract data from these tables, data analysts use queries. To learn more about that, explore BigQuery in action:

1. Log in to [BigQuery](#) and go to your console. You should find the Welcome to your SQL Workspace! landing page open. Select COMPOSE A NEW QUERY In the Bigquery console. Make sure that no tabs are open so that the entire workspace is displayed, including the Explorer pane.
2. Enter sunroof in the search bar. In the search results, expand `sunroof_solar` and then select the `solar_potential_by_postal_code` dataset.
3. Observe the Schema tab of the Explorer pane to explore the table fields.
4. Select the Preview tab to view the regions, states, yearly sunlight, and more.



Example 2: Writing a query

In order to view the entire dataset, you will need to write a query.

1. The first step is finding out the complete, correct path to the table you want to work with. Select the ellipses (three vertical dots) by the dataset `solar_potential_by_postal_code`, then select Query. A new tab will populate on your screen. Select the tab. The path to the table should be written inside two backticks.
2. Select the full path by highlighting the text including the backticks and copy it.
(Note: You can also get the full path to the project, database, and table directly by

clicking the ellipses next to the table's name in the Explorer panel on the left and selecting Copy ID.)

3. Now, click on the plus sign to create a new query. Notice that BigQuery doesn't automatically generate a **SELECT** statement in this window. Enter **SELECT** and add a space after it.
4. Put an asterisk ***** after **SELECT** to indicate you want to return the entire dataset. The asterisk lets the database know to include all columns. Without this shortcut, you would have to manually enter every column name!
5. Next, press the Enter/Return key and enter **FROM** on the second line. **FROM** indicates where the data is coming from. After **FROM**, add another space.
6. Paste in the path to the table that you copied earlier. It will read
``bigquery-public-data.sunroof_solar.solar_potential_by_postal_code``
7. Execute the query by selecting the RUN button.

The screenshot shows a Google Cloud BigQuery console interface. At the top, there's a browser tab for 'solar_potential_by_postal... ode' and a query editor titled '*Untitled'. The query editor contains two lines of SQL: 'SELECT *' and 'FROM `bigquery-public-data.sunroof\_solar.solar\_potential\_by\_postal\_code`'. Below the query editor, there are buttons for 'RUN', 'SAVE', 'SHARE', 'SCHEDULE', and 'MORE'. A 'Query' status indicator is on the right. Below the query editor, there's a 'Query results' section with a 'SAVE RESULTS' button and an 'EXPLORE DATA' link. The results are displayed in a table with columns: 'Row', 'region_name', 'state_name', 'lat_max', 'lat_min', 'lng_max', 'lng_min', 'lat_avg', and 'lng_avg'. The table contains three rows of data for Pennsylvania postal codes 19149, 19120, and 19132. The 'PREVIEW' tab is selected in the results section.

Row	region_name	state_name	lat_max	lat_min	lng_max	lng_min	lat_avg	lng_avg
1	19149	Pennsylvania	40.0512...	40.0268795	-75.0413555	-75.0925965	40.03740444034...	-75.0657051746...
2	19120	Pennsylvania	40.0527...	40.0124032	-75.0969875	-75.1387471	40.03439829328...	-75.1208872435...
3	19132	Pennsylvania	40.0058...	39.9863578	-75.1506389	-75.1906974	39.99568149075...	-75.1696451592...

Important!

Many of the public databases on BigQuery are living records and, as such, are periodically updated with new data. Throughout this course (and others in this certificate program), if your results differ from those you encounter in videos or screenshots, there's a good chance it is due to a data refresh. You can verify when a table has been refreshed by selecting it from the Explorer panel and clicking Details. You'll find the date the table was created, when it was last modified, as well as other useful information.

solar_potential_by_postal_code		QUERY	SHARE	COPY	SNAPSHOT	DELETE	EXPORT
SCHEMA	DETAILS	PREVIEW	TABLE EXPLORER	LINEAGE	DATA PROFILE	DATA QUALITY	
Table info							EDIT DETAILS
Table ID	bigquery-public-data.sunroof_solar.solar_potential_by_postal_code						
Created	Nov 16, 2017, 9:26:36 AM UTC-8						
Last modified	Nov 12, 2021, 1:42:23 PM UTC-8						
Table expiration	NEVER						
Data location	US						
Default collation							
Default rounding mode	ROUNDING_MODE_UNSPECIFIED						
Case insensitive	false						
Description	Sunroof Solar Potential By Postal Code						
Labels							
Primary key(s)							
Tags							

Example 3: Use SQL to view a piece of data

If the project doesn't require every field to be completed, you can use SQL to see a particular piece, or pieces, of data. To do this, specify a certain column name in the query.

1. For example, you might only need data from Pennsylvania. You'd begin your query the same way you just did in the previous examples: Click on the plus sign, enter **SELECT**, add a space, an asterisk (*), and then press Enter/Return.
2. Enter **FROM** and then paste ``bigquery-public-data.sunroof_solar.solar_potential_by_postal_code``. Press Enter/Return.
3. This time, add **WHERE**. It will be on the same line as the **FROM** statement. Add a space and enter `state_name` with a space before state and a space after name. `state_name` is a column name in the table.

4. Because you only want data from Pennsylvania, add `=` and `'Pennsylvania'` on the same line as `state_name`. In SQL, single quotes represent the beginning and ending of a string.
5. Execute the query with the RUN button.
6. Review the data on solar potential for Pennsylvania. Scroll through the query results.

Keep in mind that SQL queries can be written in a lot of different ways and still return the same results. You might discover other ways to write these queries!